Aaron Havener

havener.aaron@outlook.com

(727) 408-0612

Educational CVT Robot For Haptic Teleoperation

Project: Design an autonomous robot to engage students in robotics education while demonstrating the potential of continuously variable transmissions (CVTs) for teleoperation in haptic robotics

Delivered: Proof-of-concept robotic system including controller derivation and tuning, motion simulation, mechanical design, material selection, procurement, and technical documentation

Major Design Challenges: Nonlinear system dynamics complicated controller design, determining the required wheel/motor positions to achieve desired two-dimensional end-effector coordinates, and integrating sensors with the mechanical design to accurately measure desired versus actual wheel positions

Validation: Feasibility was demonstrated through controller performance, wheel-position tracking, end-effector trajectory visualization, and agreement between expected system kinematics and simulated end-effector motion

Future Work: Complete hardware integration, finalize the prototype, and experimentally validate end-effector motion accuracy

Project Resources:

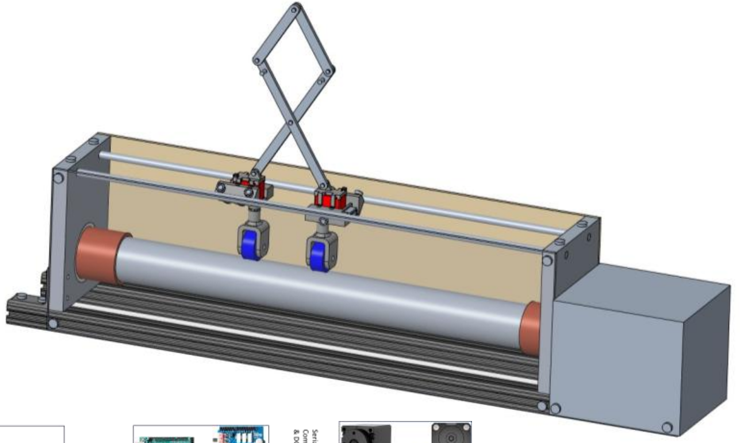
Website (best viewed on desktop)

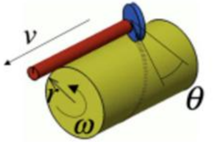
Senior Design Day Presentation

Senior Design Day Poster

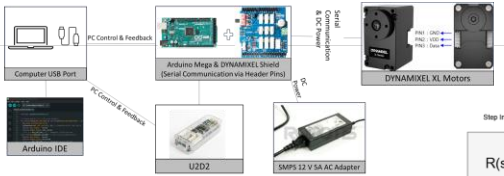
Research

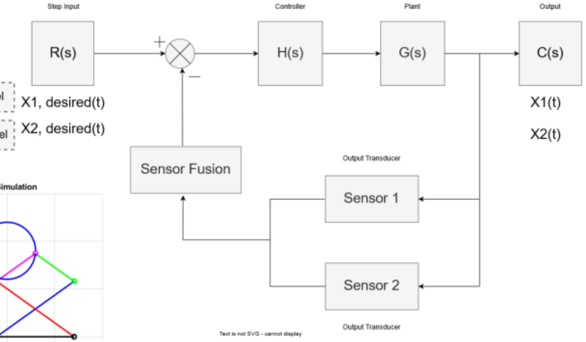
CVT Robot Code (Note: some code was commented out for the script to publish successfully)

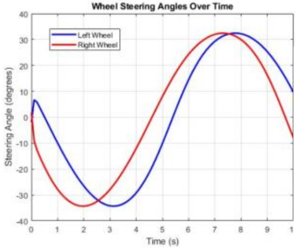


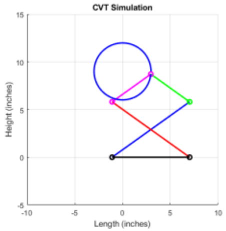


$$v = r\omega \tan\theta$$









Example Simulation Video:
https://www.youtube.com/watch?v=SCf1pLmJTnE

Aaron Havener havener.aaron@outlook.com (727) 408-0612

Ball-and-Beam Feedback Control System

Project: Design an open-loop or closed-loop control system that includes a simulation of what it does and how/why it is successful

Delivered: A PD-controlled ball-balancing simulation with dynamic visualization, transfer-function modeling, and closed-loop performance analysis

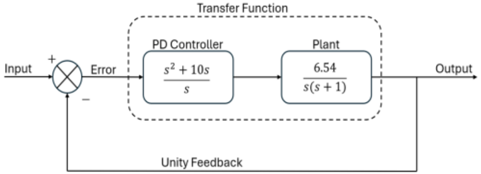
Major Design Challenges: Modeling nonlinear system dynamics, selecting stable controller gains, and incorporating closed-loop feedback for accurate position regulation

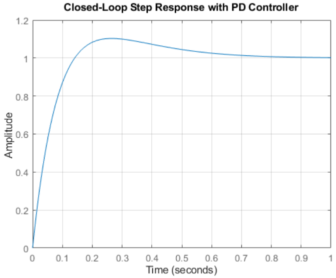
Validation: Verified controller stability by comparing open- and closed-loop responses, evaluating controller gain selection, analyzing transient performance through simulation, and visualizing system response via animation

Future Work: Further tune gains for a slower response to improve animation and for implementation on physical hardware

Project Resources:
[Project Code]()
[Final Report]()
[Presentation]()

Simulation Videos:
https://www.youtube.com/watch?v=NZizfFtlKTQ
https://www.youtube.com/watch?v=UxAXXM9ae4s





Autonomous Nonholonomic Mobile Robot

Project: Designed and implemented autonomous navigation software for a nonholonomic mobile robot using LiDAR-based mapping, path planning, and Pure Pursuit control

Delivered: Autonomous navigation software enabling waypoint following, obstacle avoidance, and real-time path planning on a physical mobile robot

Major Design Challenges: Developing Pure Pursuit steering logic, generating collision-free paths from LiDAR data, and integrating perception with autonomous navigation

Validation: Demonstrated autonomous waypoint following and obstacle avoidance through real-world testing in a constrained indoor environment

Future Work: Improve path smoothness/reduce unnecessary steering corrections and optimize navigation efficiency in dynamic environments

Test Results Video Link(s): https://www.youtube.com/shorts/MYD92z-kxPc
https://www.youtube.com/shorts/ahSgRSu2JDg